Final Competition

CS 360 Spring 2025

Task: Representation Learning

- This is also the essence of deep learning.
- Note that at the final hidden layer, we essentially transform the input data into a vector, consisting of the activations of every neuron in that layer.
- The final prediction is just a linear model using that vector as the input features.

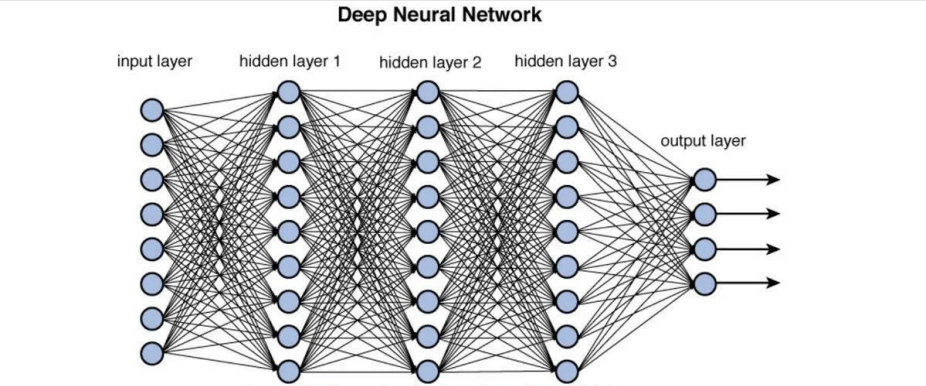
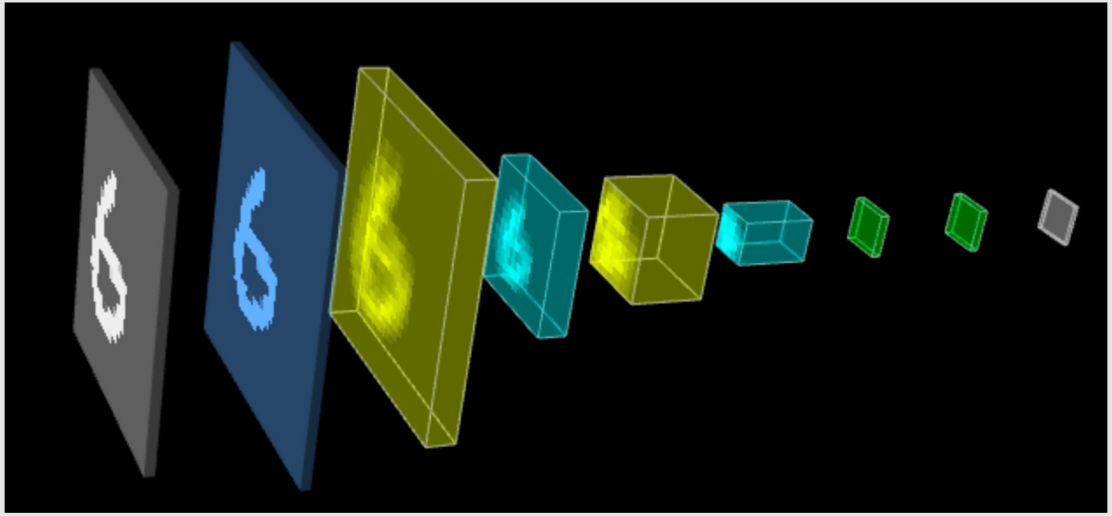


Figure 12.2 Deep network architecture with multiple layers.

Task: Representation Learning

- This allows us to learn a representation of the original data point as a vector. Vectors are easy to work with for many machine learning tasks.



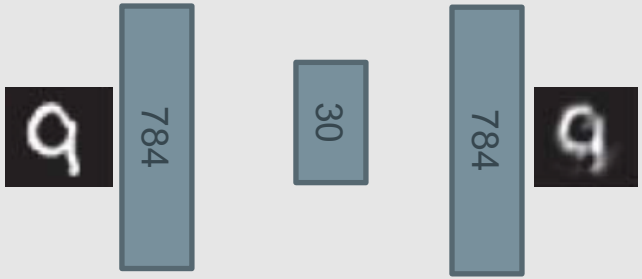
A CNN example. Play with it at: https://tensorspace.org/html/playground/lenet.html

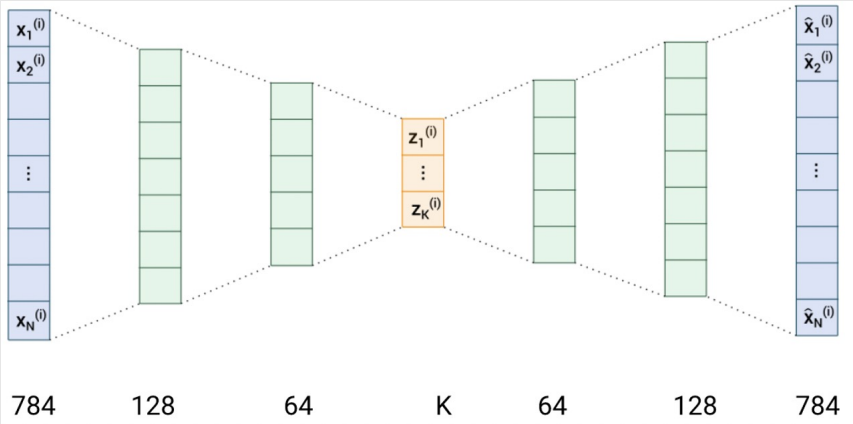
Deep Autoencoder

A special kind of deep learning architecture that learns a representation of reduced dimension.

Original Image

Deep Auto-encoder





Data: Pokemon Fusion Images

- Around 18k images given to you for training.
- You train your model (an autoencoder model), and submit to our server through KIWI. The testing data will be hidden from you.
- We will test your model on three tasks:
    - Reconstruction
    - Linear probing of the bottleneck representation
    - Sampling

Reconstruction

- Your autoencoder model has two components: encoder and decoder.
- Encoder: input data point x -> bottleneck representation z
- Decoder: bottleneck representation z -> reconstructed data point x’

- The bottleneck size z is of a limited dimension (8192).
- We can think of it as a compression of the original data.
- We will test the MSE between x and x’. I.e., there should be minimal information loss during the compression

Linear Probing of z

- Does the bottleneck feature contain valuable information?
- In addition to the images, we will also give a label of each fusion pokemon, and the label is the type (e.g., grass and ice, fire, bug and flying, etc.)
- Given z, we will train a logistic regression model on it to predict the label.
- There are 170 number of types.

Sampling

- This will not used for scoring. Only for fun.
- We will compute the mean and std over all the z generated by your model on our testing data.
- We will then sample z’ based on the mean and std.
- We then run decoder on z’ to generate a new fusion Pokemon image.
- We will display your images on the leaderboard (but not your uploaded image yet).

Scoring

- Score = reconstruction MSE / probing acc
- Leaderboard ranked by Score

- Your grade will be calculated by an equation based on the score (details released later).
- The logic is essentially: we will set up two thresholds towards the end of the competition. One baseline and one full-score line. Baseline gets 60% and full-score line gets 100%. There will be an interpolation between the two lines using some equation.

Constraints

- Both your model size and the running time (inference on our testing data points) will be constrained.
- Currently we are testing it, and the range 50M for the model size and 50 secs for the running time.

- Please don't intentionally break the competition platform on KIWI. If we find it, we will give 0 to this assignment.

To start with ...

- Search for convolutional autoencoder and find some open repos.
- Use google colab for GPU.

- Convolution is just a type of neural network layer with sparsity patterns.